MULTIMEDIA UNIVERSITY

CSE6224-SOFTWARE REQUIREMENTS ENG

Project 2 (50%)

Title: Campus Event Check-in System

Prepared By:

ID	Name	Contribution
1211305566	Alkatheri, Suliman Ali Mahdi	25%
1221101391	LIM AI NEE	25%
1221100800	MUHAMMAD AZHAR BIN AHMAD ROSEHAIZAT	25%
1211103488	YOUSEF ELSAYED EZZAT ELSAYED	25%

Group: 1

Supervision By:

Dr. Zarina Binti Che Embi (zarina.embi@mmu.edu.my)

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1. Introduction

1.1 Purpose

The purpose of this project is to develop a digital check-in and payment system for campus events that seamlessly integrates with the university's student identification database and a secure payment gateway. The system aims to automate event attendance through student ID verification and ticket validation, reducing manual processes and human error. Additionally, it enables secure on-site purchases—such as food, merchandise, and services—and equips event organizers with real-time analytics and reports to improve event management. Ultimately, the system enhances operational efficiency, security, and user experience for both students and staff.

1.2 Scope

The system allows students and vendors to check in to events via QR code, providing a streamlined and user-friendly experience. Key features include real-time notifications, integrated payment processing, and an event rating mechanism. The system does not include support for mapping, third-party navigation systems, or outdoor location services.

1.3 Product Overview

The digital check-in system is designed to manage campus event attendance and transactions efficiently. It enables students to check in using their university-issued IDs and supports both digital and physical ticket validation. The system interfaces with the university's student database for real-time identity verification and with payment gateways for secure transactions. Admin users can oversee events, monitor attendance, manage ticketing, and access comprehensive reports. The platform is intended for use by students, vendors, and administrators.

1.3.1 Product Perspective

The Campus Check-in System operates as a centralized event management platform that interacts with multiple user roles and external systems. The system interfaces with:

- Students, who can log in, register for events, view event details, and make payments.
- Vendors, who can log in, accept payments and view sales data.
- Admins, who manage events, monitor attendance, add vendor accounts, and generate reports.
- External systems, including:
    - University student database for verification,
    - Online payment gateways for transaction processing,

This interaction is illustrated in **Figure 1. 1**, which shows the high-level context of the system and its key data exchanges.

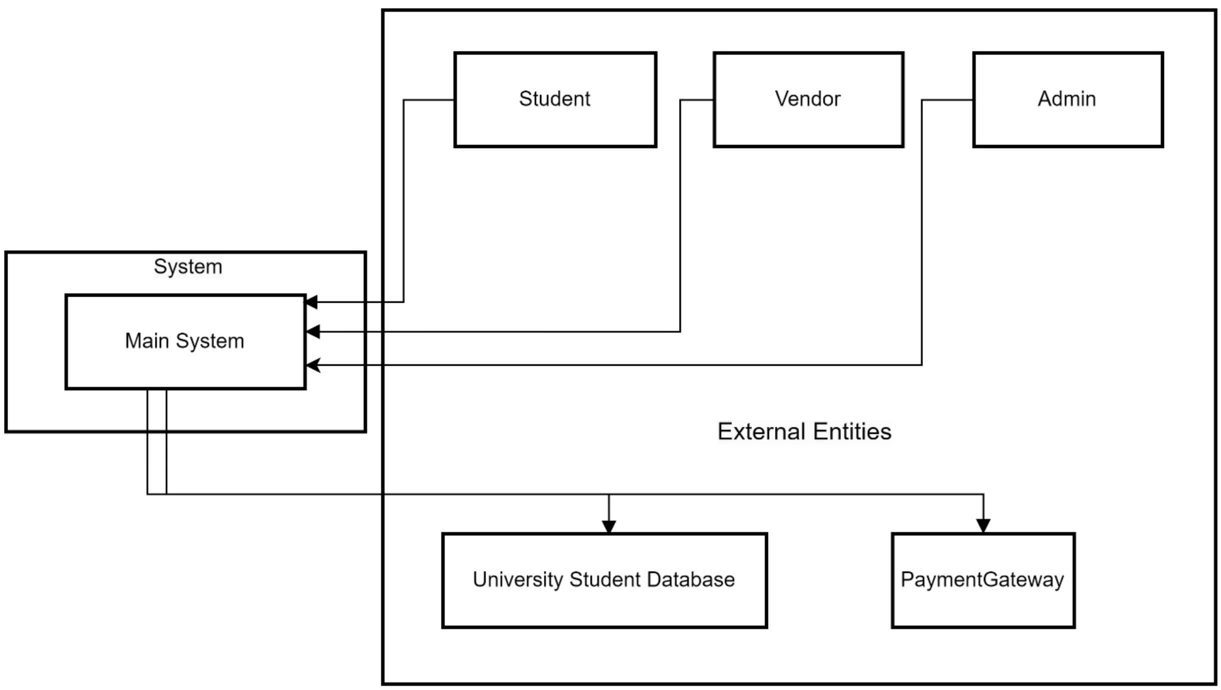

Figure 1. 1 System Context Diagram

1.3.1.1 System Goals

Table 1. 1 outlines the high-level system goals for the Campus Event Check-in & Payment System. Each goal is stated in measurable terms to guide the development team in meeting performance expectations. These objectives address critical aspects such as check-in efficiency, digital payment enablement, real-time data visibility, data accuracy, system scalability, security

compliance, and administrative workload reduction—ensuring the system delivers a seamless and impactful user experience.

Table 1. 1 System Goals Table

Requirement ID	Description	Priority	Author
REQ_GO01	Reduce average student check-in time to < 5 seconds per attendee, compared with manual paper sign-in.	High	Suliman
REQ_GO02	Enable 100 % of on-site vendor purchases to be completed digitally (QR / wallet / card) without cash handling.	High	Suliman
REQ_GO03	Provide organisers a live dashboard showing ≤ 30-second latency from check-in or purchase event to dashboard display.	Medium	Suliman
REQ_GO04	Achieve ≥ 99.5 % accuracy between recorded check-ins and ticket registrations (mismatches < 0.5 %).	High	Suliman
REQ_GO05	Support 300 concurrent users and 3 000 total event participants with no perceptible performance degradation.	High	Suliman
REQ_GO06	Maintain zero known data-breach incidents and comply with university privacy policy plus PCI DSS for all payments.	Medium	Suliman
REQ_GO07	Cut post-event attendance-report preparation time for admins by ≥ 80 % (from hours to under 10 minutes).	Medium	Suliman

1.3.1.2 System Interface

This section outlines the system interface requirements that define how the system will interact with external components and services. It specifies the integration points with institutional databases, third-party payment gateways, and backend services. These interfaces ensure secure identity validation, payment processing, data synchronization for admin functionalities, and system activity logging, thereby enabling seamless interaction between different modules and ensuring compliance, security, and real-time performance. **Table 1. 2** shows the system interface.

Table 1. 2 System Interface Requirements

Requirement ID	Description	Priority	Author
REQ_SI001	The system must be linked to the university's student database via an authenticated API or secure connection to validate student identity in real-time.	High	Suliman
REQ_SI002	The system will integrate with Stripe, a PCI-compliant digital payment service, to allow vendors to process purchases using debit, credit, or e-wallets.	High	Suliman
REQ_SI003	Admin interfaces must connect to the centralized backend to retrieve and update data related to events, tickets, and attendance.	Medium	Suliman
REQ_SI004	All check-ins, purchases, and admin actions will be logged securely.	High	Suliman

1.3.1.3 User Interface

This section specifies the logical characteristics of each interface between the software product and its end users as shown in **Table 1. 3**. It describes how the user interface should appear and behave from the perspective of the user. These requirements ensure that the system remains accessible, responsive, and user-friendly across different devices. It emphasizes consistent layout and interaction patterns, provides clear feedback mechanisms, supports multilingual needs, and accommodates accessibility standards for an inclusive user experience.

Table 1. 3 User Interface Requirements

Requirement ID	Description	Priority	Author
REQ_UI001	The system shall offer a responsive design that adjusts to different screen sizes and devices.	High	Suliman
REQ_UI002	Interfaces must maintain consistency in layout, colors, icons, and interaction patterns to support usability.	High	Suliman
REQ_UI003	Navigation must be simple and intuitive, allowing users to complete tasks with minimal steps.	High	Suliman
REQ_UI004	All forms shall provide real-time validation feedback, including helpful error messages for incorrect input.	Medium	Suliman
REQ_UI005	Visual feedback (e.g., success, loading, or failure indicators) must be provided for all user actions.	Medium	Suliman
REQ_UI006	Text fields, buttons, tables, and other components must be clearly labeled and accessible.	High	Suliman
REQ_UI007	QR code-based interactions (for check-in and vendor payments) must be integrated seamlessly into the mobile interface.	High	Suliman
REQ_UI008	The UI must be accessible, supporting standard keyboard navigation and readable contrast ratios.	High	Suliman
REQ_UI009	The system should allow for language localization in future versions to support multilingual users.	Low	Suliman

1.3.1.4 Hardware Interface

This section describes the physical and technological hardware requirements necessary for the system to operate effectively, as outlined in **Table 1. 4.** It includes specifications for the types of devices to be used by students, vendors, and administrators, covering essential features such as internet connectivity, camera functionality, QR code scanning, and display resolution. It also outlines the infrastructure needed to support secure communication and reliable server hosting to ensure consistent performance and data protection throughout system operations.

Table 1. 4 Hardware Interface Requirements

Requirement ID	Description	Priority	Author
REQ_HW001	Student devices must be smartphones (Android/iOS) equipped with cameras to enable QR scanning at event check-in.	High	Suliman
REQ_HW002	Student devices must be internet-enabled, supporting either Wi-Fi or mobile data connections.	High	Suliman
REQ_HW003	The app will access the camera for QR scanning functionality on student devices.	High	Suliman
REQ_HW004	Vendor devices must be smartphones or tablets capable of managing sales and logging payments.	High	Suliman
REQ_HW005	Vendor devices must have reliable internet connectivity to ensure real-time transaction logging.	High	Suliman
REQ_HW006	Admins will use laptops or desktops to manage events, generate reports, and control attendance.	High	Suliman
REQ_HW007	Admin devices must support standard web browsers and secure system access protocols.	High	Suliman
REQ_HW008	Admins may use tablets or laptops to display QR codes for check-in purposes.	Medium	Suliman
REQ_HW009	QR code display devices should have high-resolution displays to ensure easy and accurate scanning.	Medium	Suliman
REQ_HW010	The backend system must be hosted on a university-managed server infrastructure.	High	Suliman
REQ_HW011	All client-server communications must be secured using HTTPS with TLS encryption.	High	Suliman

1.3.1.5 Software Interface

This section defines the software-level interactions between the system and external services, as illustrated in **Table 1. 5.** It outlines how the system will securely connect to university databases and third-party services such as Stripe and the Elastic Stack. These interfaces facilitate operations like login verification, payment processing, real-time QR code generation, and backend activity logging. The requirements ensure secure communication using APIs and encryption, while also supporting data analysis through integrated monitoring tools like Logstash and Kibana.

Table 1. 5 Software Interface Requirements

Requirement ID	Description	Priority	Author
REQ_SW001	The system shall connect to the university's student information system via a hosted API to verify credentials during login and check-in.	High	Suliman
REQ_SW002	The system shall integrate with Stripe as a payment gateway to manage payment sessions, confirm transactions, and verify payment success.	High	Suliman
REQ_SW003	Communication with Stripe shall occur over HTTPS, with API keys securely stored in the backend system.	High	Suliman
REQ_SW004	The system shall use the qr_flutter library or an equivalent package to generate dynamic QR codes for each event.	Medium	Suliman
REQ_SW005	Generated QR codes shall include encrypted event IDs and be displayed in real-time during check-in.	High	Suliman
REQ_SW006	The system shall utilize the Elastic Stack (Elasticsearch, Logstash, Kibana) for backend logging of check-ins, transactions, and admin activities.	Medium	Suliman
REQ_SW007	All logs shall be pushed to Logstash and visualized through Kibana dashboards for monitoring and analysis.	Medium	Suliman

1.3.1.6 Communication Interfaces

This section outlines the protocols and methods used for data exchange between the system and its components, as presented in **Table 1. 6**. It focuses on ensuring secure and encrypted communication across all channels, including interactions with the university database, payment gateway, internal system modules, and user-facing interfaces. All communications are enforced through HTTPS using TLS encryption to maintain data confidentiality, integrity, and security throughout system operations.

Table 1. 6 Communication Interface Requirements

Requirement ID	Description	Priority	Author
REQ_COM001	Communication with the student database shall use HTTPS over a REST API to securely validate student records during login and check-in.	High	Suliman
REQ_COM002	Payment transactions shall be transmitted securely using HTTPS with TLS encryption between student devices and the Stripe system.	High	Suliman
REQ_COM003	Internal system communications, including QR code generation, event syncing, and logging, shall use HTTPS to ensure secure operations.	High	Suliman
REQ_COM004	All frontend interfaces for students, vendors, and admins shall communicate with the backend over HTTPS via web-based protocols to ensure secure and encrypted interactions.	High	Suliman

1.3.2 Product Functions

Table 1. 7 provides an overview of the essential functions offered by the system. Each function represents a key capability designed to support core event operations such as student check-in, ticket verification, payment processing, and attendance tracking. The descriptions highlight how these features interact with various components, including the student database and reporting tools, to ensure a secure, efficient, and insightful event experience for students, vendors, and administrators.

Table 1. 7 Product Functions Table

Function	Description
Student Check-in	Allows students to check in to events using their university ID cards or student numbers.
Ticket Verification	Validates tickets during entry to ensure that only authorized participants are admitted.
Payment Processing	Enables secure on-site purchases (e.g., food, merchandise) through integration with payment gateways.
Real-time Attendance Tracking	Records attendance as students check in and provides up-to-date data to the admin interface.
Student Database Integration	Connects to the university's identification database for real-time verification of student status.
Analytics and Reports	Generates post-event reports including attendance logs, transaction summaries, and ticket scan histories for students and vendors.

1.3.3 User Characteristics

Table 1. 8 outlines the primary user roles that will interact with the system. Each role is accompanied by a brief description of their expected usage and the level of knowledge required to operate the system effectively. This helps ensure the system design supports users with varying technical backgrounds, from students performing simple check-ins to administrators and vendors managing transactions and reports.

Table 1. 8 User Characteristics Table

Role	Description	Required Knowledge
Student	University students using the system to check in and make purchases at events.	Basic familiarity with mobile or web-based applications.
Admin	Staff responsible for managing events, tracking attendance, and accessing reports.	Familiarity with admin dashboards, event management tools, and reporting interfaces.
Vendor	On-site service providers (e.g., food or merchandise vendors) using the system to process transactions and monitor sales.	Basic understanding of digital payment platforms and ability to navigate the vendor interface.

1.3.4 Limitations

- **Dependence on Internet Connectivity**: The system requires a stable internet connection for real-time check-ins, payment processing, and data synchronization.
- **Scalability Constraints**: Performance may degrade in large-scale events without proper server infrastructure.
- **Lack of Offline Support**: Version 1.0 does not support offline operations, including check-in and payment processing.
- **User Training Needs**: Admins and vendors may require minimal training to operate the system efficiently and avoid errors.
- **Device Compatibility**: Admins and vendors must use compatible tablets or laptops; inadequate hardware could hinder functionality.

1.4 Definitions

This section provides definitions of key terms used throughout the document, as shown in **Table 1. 9**. These definitions help establish a clear and consistent understanding of core concepts such as integration, analytics, check-in systems, payment gateways, and user roles. Defining these terms ensures that all stakeholders share a common interpretation of system components, features, and interactions.

Table 1. 9 Definition of Terms Table

Term	Definition
Integration	The process of linking the system with external platforms such as databases and payment gateways.
Analytics	Data analysis activities aimed at extracting insights, such as user activity or attendance patterns.
On-site Purchases	Purchases made at the event location (e.g., food, merchandise).
Payment	A service that securely processes digital payments such as credit/debit card

Gateway	transactions.
Check-in System	A digital tool used to verify participant presence at events.
Ticket Verification	The validation process that determines whether an event ticket is genuine and valid.
End Users	Individuals interacting directly with the system (students, admins, vendors).
Backend	The server-side logic and infrastructure that supports data processing, storage, and core functionality.

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3. Requirements

3.1 External Interfaces

This section defines the external interface requirements of the Campus Event Check-In System. It describes the user-facing actions, and the corresponding inputs and outputs expected in each interaction. The interface elements are designed to offer a seamless and user-friendly experience, while ensuring secure and efficient communication with external components such as the university's student database, integrated payment systems, and QR scanning tools. These interfaces serve as the connection point between users and the system's core functionalities.

3.1.1 Login

Tables **Table 3.1.1 – Table 3.1.4** define the interface components used in the login user interface.

Table 3.1. 1 REQ_IO0001 Table

Requirement ID	REQ_IO0001	**Version**	1.0
Item	Login Button (Input)		
Item Description	A button labeled "Login"		
Item Purpose	Submits login credentials for authentication		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0002, REQ_IO0003 (Both fields must be filled)		
Author	Sulaiman		

Table 3.1. 2 REQ_IO0002 Table

Requirement ID	REQ_IO0002	**Version**	1.0
Item	Student ID Field (Input)		
Item Description	A text field labeled "Student ID"		
Item Purpose	Allow users to enter their Student ID		
Input Format	String	**Valid Input**	ASCII code from decimal 32 to 126
Related I/O	None		
Author	Sulaiman		

Table 3.1. 3 REQ_IO0003 Table

Requirement ID	REQ_IO0003	**Version**	1.0
Item	Password Field (Input)		
Item Description	A text field labeled "Password"		
Item Purpose	Allow users to enter their password		
Input Format	String	**Valid Input**	ASCII code from

			decimal 32 to 126
Related I/O	None		
Author	Sulaiman		

Table 3.1. 4 REQ_IO0004 Table

Requirement ID	REQ_IO0004	**Version**	1.0
Item	Failure Message (Output)		
Item Description	Toast message for login failure		
Item Purpose	Notifies users of incorrect login		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0001 (Only displayed after submission)		
Author	Sulaiman		

3.1.2 Event Registration

Tables **Table 3.1.5 – Table 3.1.10** define the interface components used in the event registration user interface.

Table 3.1. 5 REQ_IO0005 Table

Requirement ID	REQ_IO0005	**Version**	1.0
Item	Event List (Output)		
Item Description	A list displaying available events with key details		
Item Purpose	Allow students to browse and select events		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0006		
Author	Sulaiman		

Table 3.1. 6 REQ_IO0006 Table

Requirement ID	REQ_IO0006	**Version**	1.0
Item	Register Button (Input)		
Item Description	Button labeled “Register” next to each event		
Item Purpose	Initiates the registration process for a selected event		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0005, REQ_IO0007		
Author	Sulaiman		

Table 3.1. 7 REQ_IO0007 Table

Requirement ID	REQ_IO0007	**Version**	1.0
Item	Event Details Modal (Output)		

Item Description	A popup/modal showing full event information		
Item Purpose	Allow students to review event details before confirming		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0006, REQ_IO0008		
Author	Sulaiman		

Table 3.1. 8 REQ_IO0008 Table

Requirement ID	REQ_IO0008	**Version**	1.0
Item	Confirm Registration Button (Input)		
Item Description	Button to finalize event registration		
Item Purpose	Submits event registration request		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0007		
Author	Sulaiman		

Table 3.1. 9 REQ_IO0009 Table

Requirement ID	REQ_IO0009	**Version**	1.0
Item	Registration Success Message (Output)		
Item Description	A confirmation message after successful registration		
Item Purpose	Informs student that registration is complete		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0008		
Author	Sulaiman		

Table 3.1. 10 REQ_IO0010 Table

Requirement ID	REQ_IO0010	**Version**	1.0
Item	Registration Error Message (Output)		
Item Description	A message displayed when registration fails		
Item Purpose	Notifies student of issues with registration		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0008		
Author	Sulaiman		

3.1.3 Event Check-in

Tables **Table 3.1.11 – Table 3.1.14** define the interface components used in the event check-in user interface.

Table 3.1. 11 REQ_IO0011 Table

Requirement ID	REQ_IO0011	**Version**	1.0
Item	QR Scanner Button (Input)		
Item Description	A button that activates the device camera for QR scanning		
Item Purpose	Allow student to initiate the check-in process by scanning a QR code		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0012		
Author	Sulaiman		

Table 3.1. 12 REQ_IO0012 Table

Requirement ID	REQ_IO0012	**Version**	1.0
Item	QR Code Data (Input)		
Item Description	Encoded event-specific QR code content		
Item Purpose	Provides event identity to validate check-in		
Input Format	String	**Valid Input**	Valid event token (UUID format or predefined code)
Related I/O	REQ_IO0011		
Author	Sulaiman		

Table 3.1. 13 REQ_IO0013 Table

Requirement ID	REQ_IO0013	**Version**	1.0
Item	Check-in Confirmation Message (Output)		
Item Description	A success message upon successful check-in		
Item Purpose	Confirms successful attendance logging		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0011		
Author	Sulaiman		

Table 3.1. 14 REQ_IO0014 Table

Requirement ID	REQ_IO0014	**Version**	1.0
Item	Error Message (Output)		
Item Description	A message displayed if QR is invalid or student not registered		
Item Purpose	Informs user that check-in failed		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0011		
Author	Sulaiman		

3.1.4 Make Payment

Tables **Table 3.1.15 – Table 3.1.20** define the interface components used in the make payment user interface.

Table 3.1. 15 REQ_IO0015 Table

Requirement ID	REQ_IO0015	**Version**	1.0
Item	QR Scanner Button (Input)		
Item Description	A button to activate QR code scanner to identify the vendor		
Item Purpose	A button to activate QR code scanner to identify the vendor		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0016		
Author	Lim Ai Nee		

Table 3.1. 16 REQ_IO0016 Table

Requirement ID	REQ_IO0016	**Version**	1.0
Item	Vendor QR Code (Input)		
Item Description	Encoded vendor ID or payment session token		
Item Purpose	Identifies which vendor the payment is intended for		
Input Format	String	**Valid Input**	Valid vendor ID or session token
Related I/O	REQ_IO0015		
Author	Lim Ai Nee		

Table 3.1. 17 REQ_IO0017 Table

Requirement ID	REQ_IO0017	**Version**	1.0
Item	Amount Field (Input)		
Item Description	A numeric text field for entering payment amount		
Item Purpose	A numeric text field for entering payment amount		
Input Format	Number	**Valid Input**	Positive decimal number (e.g., 1.00 to 999.99)
Related I/O	REQ_IO0018		
Author	Lim Ai Nee		

Table 3.1. 18 REQ_IO0018 Table

Requirement ID	REQ_IO0018	**Version**	1.0
Item	Pay Button (Input)		
Item Description	A button to submit payment request		
Item Purpose	Confirms and initiates the payment		

Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0017		
Author	Lim Ai Nee		

Table 3.1. 19 REQ_IO0019 Table

Requirement ID	REQ_IO0019	**Version**	1.0
Item	Payment Confirmation Message (Output)		
Item Description	A success message displayed after payment completion		
Item Purpose	Confirms that the payment has been processed		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0018		
Author	Lim Ai Nee		

Table 3.1. 20 REQ_IO0020 Table

Requirement ID	REQ_IO0020	**Version**	1.0
Item	Payment Error Message (Output)		
Item Description	Message shown when payment fails		
Item Purpose	Alerts user to retry or check their input		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_ IO0018		
Author	Lim Ai Nee		

3.1.5 Manage Events

Tables **Table 3.1.21 – Table 3.1.29** define the interface components used in the manage events user interface.

Table 3.1. 21 REQ_IO0021 Table

Requirement ID	REQ_IO0021	**Version**	1.0
Item	Event Table (Output)		
Item Description	A table displaying all existing events with action buttons		
Item Purpose	Allows admins to view, edit, or delete existing events		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0022, REQ_IO0023, REQ_IO0024		
Author	Lim Ai Nee		

Table 3.1. 22 REQ_IO0022 Table

Requirement ID	REQ_IO0022	**Version**	1.0
Item	Create Button (Input)		

Item Description	A button labeled “Create Event”		
Item Purpose	Opens a form/modal to create a new event		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0025 to REQ_IO0029		
Author	Lim Ai Nee		

Table 3.1. 23 REQ_IO0023 Table

Requirement ID	REQ_IO0023	**Version**	1.0
Item	Edit Button (Input)		
Item Description	A button next to each event in the table to edit it		
Item Purpose	Opens the form pre-filled with event details for editing		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0025 to REQ_IO0029		
Author	Lim Ai Nee		

Table 3.1. 24 REQ_IO0024 Table

Requirement ID	REQ_IO0024	**Version**	1.0
Item	Delete Button (Input)		
Item Description	A button next to each event for deletion		
Item Purpose	Deletes the selected event after confirmation		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	None		
Author	Lim Ai Nee		

Table 3.1. 25 REQ_IO0025 Table

Requirement ID	REQ_IO0025	**Version**	1.0
Item	Event Name Field (Input)		
Item Description	Text field to enter the event name		
Item Purpose	Specifies the name of the event		
Input Format	String	**Valid Input**	Alphabetic + alphanumeric string (3–50 characters)
Related I/O	REQ_IO0022, REQ_IO0023		
Author	Lim Ai Nee		

Table 3.1. 26 REQ_IO0026 Table

Requirement ID	REQ_IO0026	**Version**	1.0
Item	Event Date Field (Input)		
Item Description	Date picker for selecting the event date		
Item Purpose	Specifies when the event will occur		

Input Format	Date	**Valid Input**	Any valid future date
Related I/O	REQ_IO0022, REQ_IO0023		
Author	Lim Ai Nee		

Table 3.1. 27 REQ_IO0027 Table

Requirement ID	REQ_IO0027	**Version**	1.0
Item	Location Field (Input)		
Item Description	Text field to enter the location of the event		
Item Purpose	Specifies the venue where the event will take place		
Input Format	String	**Valid Input**	Free-text, max 100 characters
Related I/O	REQ_IO0022, REQ_IO0023		
Author	Lim Ai Nee		

Table 3.1. 28 REQ_IO0028 Table

Requirement ID	REQ_IO0028	**Version**	1.0
Item	Capacity Field (Input)		
Item Description	Numeric field to define max number of attendees		
Item Purpose	Limits how many students can register		
Input Format	Integer	**Valid Input**	1–1000
Related I/O	REQ_IO0022, REQ_IO0023		
Author	Lim Ai Nee		

Table 3.1. 29 REQ_IO0029 Table

Requirement ID	REQ_IO0029	**Version**	1.0
Item	Save/Update Button (Input)		
Item Description	A button to save or update event information		
Item Purpose	Commits the event changes to the system		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0025 to REQ_IO0028		
Author	Lim Ai Nee		

3.1.6 Manage Attendance

Tables **Table 3.1.30** – **Table 3.1.34** define the interface components used in the manage attendance user interface.

Table 3.1. 30 REQ_IO0030 Table

Requirement ID	REQ_IO0030	**Version**	1.0
Item	Event Dropdown (Input)		

Item Description	Dropdown list of events for selection		
Item Purpose	Allows admin to select the event for attendance management		
Input Format	Dropdown	**Valid Input**	Valid event names from system
Related I/O	REQ_IO0031, REQ_IO0032		
Author	Azhar		

Table 3.1. 31 REQ_IO0031 Table

Requirement ID	REQ_IO0031	**Version**	1.0
Item	Generate QR Button (Input)		
Item Description	A button to generate the event-specific check-in QR code		
Item Purpose	Enables the admin to display the QR code used by students to check in		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0030		
Author	Azhar		

Table 3.1. 32 REQ_IO0032 Table

Requirement ID	REQ_IO0032	**Version**	1.0
Item	QR Code Image (Output)		
Item Description	QR code generated for check-in		
Item Purpose	Displayed on the screen for scanning by students		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0031		
Author	Azhar		

Table 3.1. 33 REQ_IO0033 Table

Requirement ID	REQ_IO0033	**Version**	1.0
Item	Attendance List Table (Output)		
Item Description	Table listing all students who checked in		
Item Purpose	Allows admin to view all registered attendees in real time		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0030, REQ_IO0034		
Author	Azhar		

Table 3.1. 34 REQ_IO0034 Table

Requirement ID	REQ_IO0034	**Version**	1.0
Item	Refresh Button (Input)		
Item Description	A button to reload the attendance list		
Item Purpose	Refreshes the displayed attendance table to include latest check-ins		

Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0033		
Author	Azhar		

3.1.7 *Manage Payment*

Tables **Table 3.1.35 – Table 3.1.38** define the interface components used in the manage payment user interface.

Table 3.1. 35 REQ_IO0035 Table

Requirement ID	REQ_IO0035	**Version**	1.0
Item	Payment Amount Field (Input)		
Item Description	Text field where vendor inputs the amount to be charged		
Item Purpose	Captures the transaction amount for a vendor sale		
Input Format	Decimal Number	**Valid Input**	Positive values (e.g., 1.00 – 999.99)
Related I/O	REQ_IO0036		
Author	Azhar		

Table 3.1. 36 REQ_IO0036 Table

Requirement ID	REQ_IO0036	**Version**	1.0
Item	Confirm Payment Button (Input)		
Item Description	Button to submit the entered payment		
Item Purpose	Saves the transaction to the database		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0035		
Author	Azhar		

Table 3.1. 37 REQ_IO0037 Table

Requirement ID	REQ_IO0037	**Version**	1.0
Item	Success Message (Output)		
Item Description	Message confirming successful payment submission		
Item Purpose	Informs vendor that transaction was recorded		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0036		
Author	Azhar		

Table 3.1. 38 REQ_IO0038 Table

Requirement ID	REQ_IO0038	**Version**	1.0
Item	Error Message (Output)		

Item Description	Message shown if input is missing or invalid		
Item Purpose	Warns vendor to enter a valid amount		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0035		
Author	Azhar		

3.1.8 View Sales Report

Tables **Table 3.1.39 – Table 3.1.42** define the interface components used in the view sales report user interface.

Table 3.1. 39 REQ_IO0039 Table

Requirement ID	REQ_IO0039	**Version**	1.0
Item	Date Range Picker (Input)		
Item Description	A pair of date fields (start and end)		
Item Purpose	Allows vendor to select the time period for the report		
Input Format	Date	**Valid Input**	Valid past dates (from - to)
Related I/O	REQ_IO0040		
Author	Yousef		

Table 3.1. 40 REQ_IO0040 Table

Requirement ID	REQ_IO0040	**Version**	1.0
Item	Generate Report Button (Input)		
Item Description	Button to submit the date range and retrieve data		
Item Purpose	Triggers backend process to compile sales report		
Input Format	Button	**Valid Input**	Not Applicable
Related I/O	REQ_IO0039, REQ_IO0041, REQ_IO0042		
Author	Yousef		

Table 3.1. 41 REQ_IO0041 Table

Requirement ID	REQ_IO0041	**Version**	1.0
Item	Sales Report Table (Output)		
Item Description	Table showing list of transactions, dates, and totals		
Item Purpose	Displays results of report request		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0040		
Author	Yousef		

Table 3.1. 42 REQ_IO0042 Table

Requirement ID	REQ_IO0042	**Version**	1.0
Item	Empty Result Message (Output)		
Item Description	Message shown when no data is available in selected range		
Item Purpose	Informs vendor of absence of sales records		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0040		
Author	Yousef		

3.1.9 Generate Event Report

Tables **Table 3.1.43 – Table 3.1.46** define the interface components used in the generate event report user interface.

Table 3.1. 43 REQ_IO0043 Table

Requirement ID	REQ_IO0043	**Version**	1.0
Item	Event Dropdown (Input)		
Item Description	Dropdown menu listing past events		
Item Purpose	Allows admin to select the event for report generation		
Input Format	Dropdown	**Valid Input**	Valid past event names
Related I/O	REQ_IO0044, REQ_IO0045		
Author	Yousef		

Table 3.1. 44 REQ_IO0044 Table

Requirement ID	REQ_IO0044	**Version**	1.0
Item	Event Dropdown (Input)		
Item Description	Dropdown menu listing past events		
Item Purpose	Allows admin to select the event for report generation		
Input Format	Dropdown	**Valid Input**	Valid past event names
Related I/O	REQ_IO0044, REQ_IO0045		
Author	Yousef		

Table 3.1. 45 REQ_IO0045 Table

Requirement ID	REQ_IO0045	**Version**	1.0
Item	Event Report Table (Output)		
Item Description	Table summarizing attendance and sales for selected event		
Item Purpose	Presents full event metrics in readable format		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0044		
Author	Yousef		

Table 3.1. 46 REQ_IO0046 Table

Requirement ID	REQ_IO0046	Version	1.0
Item	No Data Message (Output)		
Item Description	Message shown when the selected event has no reportable data		
Item Purpose	Informs admin of empty report case		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	REQ_IO0044		
Author	Yousef		

3.1.10 Add Vendor Account

Tables **Table 3.1.47 – Table 3.1.51** define the interface components used in the add vendor account user interface.

Table 3.1. 47 REQ_IO0047 Table

Requirement ID	REQ_IO0047	Version	1.0
Item	Vendor Name Field (Input)		
Item Description	Free-text field for vendor's legal or trade name		
Item Purpose	Records vendor identifier for account creation		
Input Format	String	**Valid Input**	3–50 UTF-8 characters (letters, spaces, hyphen, apostrophe)
Related I/O	REQ_IO0051		
Author	Suliman		

Table 3.1. 48 REQ_IO0048 Table

Requirement ID	REQ_IO0048	Version	1.0
Item	Vendor Email Field (Input)		
Item Description	Text box requiring unique, valid e-mail address		
Item Purpose	Serves as username for vendor login; used to send credentials		
Input Format	E-mail	**Valid Input**	RFC 5322-compliant address; must not already exist
Related I/O	REQ_IO0051		
Author	Suliman		

Table 3.1. 49 REQ_IO0049 Table

Requirement ID	REQ_IO0049	Version	1.0
Item	Vendor Phone Field (Input)		
Item Description	Vendor's phone number for contact purposes		

Item Purpose	Stores vendor contact phone for support		
Input Format	Numeric string	**Valid Input**	8–15 digits
Related I/O	REQ_IO0051		
Author	Suliman		

Table 3.1. 50 REQ_IO0050 Table

Requirement ID	REQ_IO0050	**Version**	1.0
Item	Password Field (Input)		
Item Description	A text field labeled "Password"		
Item Purpose	Allow users to enter their password		
Input Format	String	**Input Format**	String
Related I/O	REQ_IO0051		
Author	Sulaiman		

Table 3.1. 51 REQ_IO0051 Table

Requirement ID	REQ_IO0051	**Version**	1.0
Item	Create Button (Input)		
Item Description	Commits new vendor record		
Item Purpose	Save the vendor account in the database		
Input Format	Not Applicable	**Valid Input**	Not Applicable
Related I/O	None		
Author	Suliman		

3.2 Functions

This section outlines the core functions of the Campus Event Check-In System. The system is designed to facilitate event participation, registration, attendance tracking, and transaction processing for university-based events. It supports three primary user roles: **Students**, **Vendors**, and **Admins**, each with distinct capabilities within the system.

3.2.1 Functional Requirements

The functional requirements of the system are organized by user role and are derived from the system's high-level use cases. These requirements define what the system shall do to support event check-in, ticket management, payment processing, and administrative oversight.

Figure 3. 1 presents the **Use Case Diagram** of the Campus Event Check-In System, illustrating the interaction between system users and functional components.

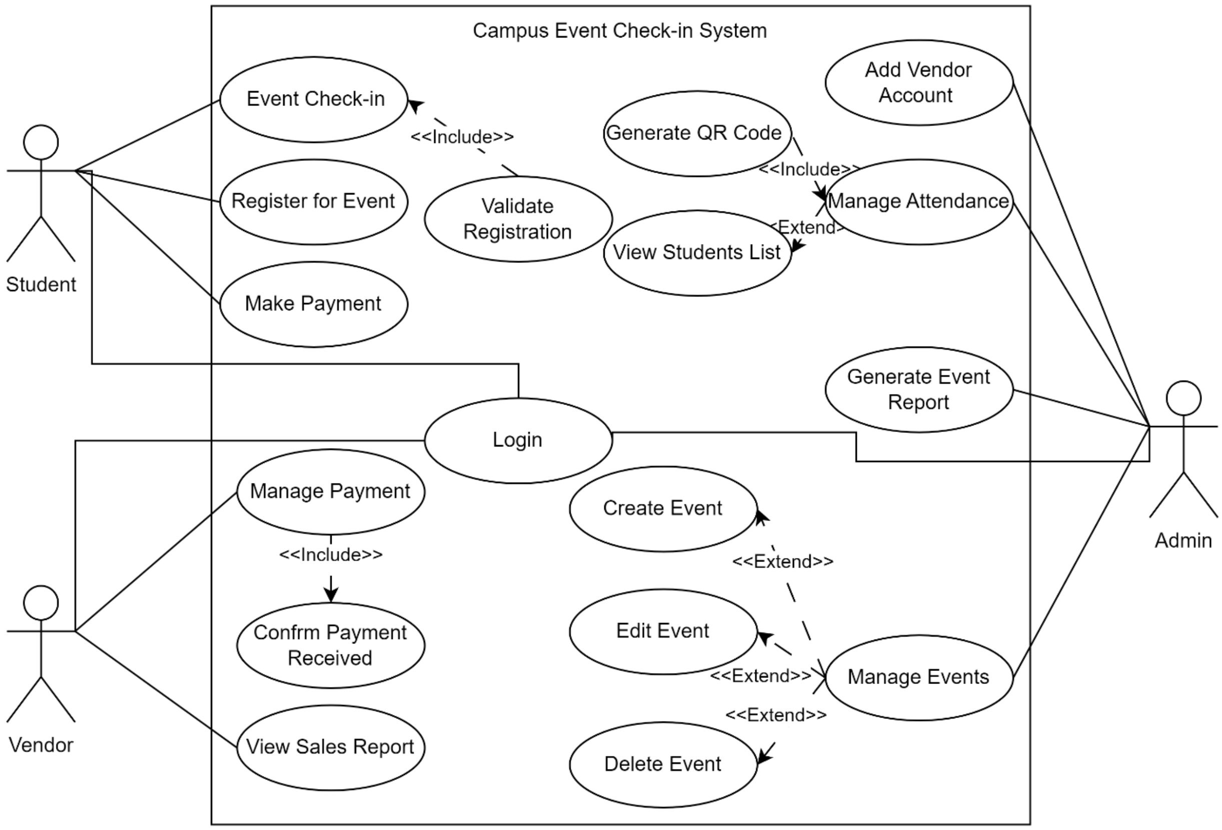

Figure 3. 1 Use Case Diagram – Campus Event Check-In System

Each use case represents a major system function. The following sub-sections (3.2.1) break down these use cases into detailed functional requirements.

3.2.1.1 Login

This section defines the functional requirements and use case specification for the **Login** feature of the system. It includes a detailed description of the system behavior, the primary actors involved, and the steps taken during the interaction. A sequence diagram is also provided to visually represent the process flow.

Table 3.2. 1 FR0101 Table

Requirement ID	**FR0101**	**Version**	**2.0**
Description	The system shall allow users (student, admin, vendor) to log in using their unique credentials.		
Author	Suliman		

Table 3.2. 2 FR0102 Table

Requirement ID	FR0102	Version	2.0
Description	The system shall validate login credentials against the registered database records.		
Author	Suliman		

Table 3.2. 3 FR0103 Table

Requirement ID	FR0103	Version	2.0
Description	The system shall display an error message if invalid credentials are entered.		
Author	Suliman		

Table 3.2. 4 FR0104 Table

Requirement ID	FR0104	Version	2.0
Description	The system shall prompt the user if required login fields are left empty.		
Author	Suliman		

Table 3.2. 5 F001 Login Use Case Specification Table

Use Case ID	UC001	Version	2.0
Feature	F001 – Login		
Purpose	To authenticate users before granting access to the system		
Actor	Student / Admin / Vendor		
Trigger	User opens the system and enters login credentials		
Precondition	User must have a valid account		
Scenario Name	Login Process		
Main Flow	1. User opens the application 2. System displays the login form 3. User enters credentials 4. System checks input format (non-empty, correct format) 5. System verifies credentials against the database 6. If valid, the user is redirected to the correct interface		
Alternate Flow – Empty Fields	4.1 System shows: “Please fill in all required fields”		
Alternate Flow – Invalid Credentials	5.1 System shows: “Invalid username or password”		
Author	Suliman		

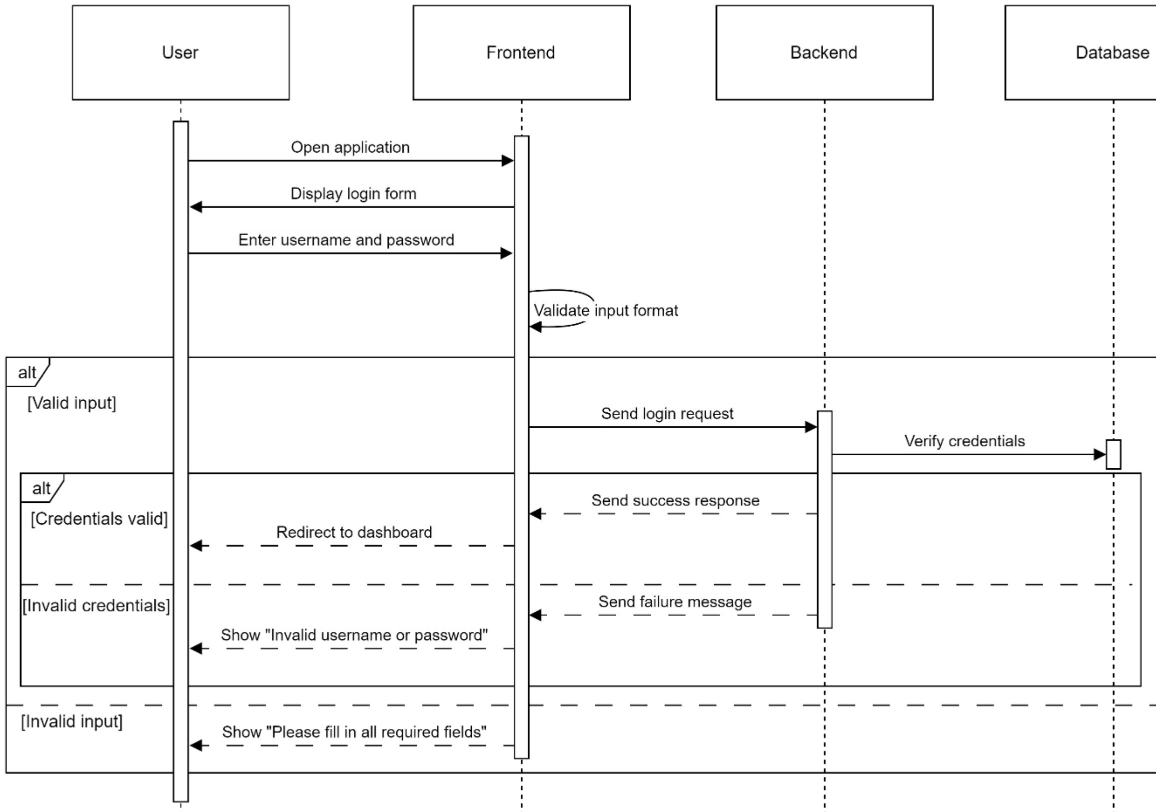

Figure 3. 2 F001 Login Sequence Diagram

3.2.1.2 Event Registration

This section defines the functional requirements and use case specification for the Event Registration feature of the system. It includes a detailed description of the system behavior, the primary actors involved, and the steps taken during the interaction. A sequence diagram is also provided to visually represent the process flow.

Table 3.2. 6 FR0201 Table

Requirement ID	**FR0201**	**Version**	**2.0**
Description	The system shall allow students to register for available events.		
Author	Suliman		

Table 3.2. 7 FR0202 Table

Requirement ID	**FR0202**	**Version**	**2.0**
Description	The system shall check ticket availability before proceeding with registration.		
Author	Suliman		

Table 3.2. 8 FR0203 Table

Requirement ID	**FR0203**	**Version**	**2.0**
Description	The system shall process the event registration payment using an integrated payment gateway.		
Author	Suliman		

Table 3.2. 9 FR0204 Table

Requirement ID	**FR0204**	**Version**	**2.0**
Description	The system shall update the student's registration record upon successful payment.		
Author	Suliman		

Table 3.2. 10 FR0205 Table

Requirement ID	**FR0205**	**Version**	**2.0**
Description	The system shall notify the student of successful registration or payment failure.		
Author	Suliman		

Table 3.2. 11 F002 – Event Registration Use Case Specification Table

Use Case ID	UC02	**Version**	2.0
Feature	F002 – Event Registration		
Purpose	To allow students to register for events and complete payment		
Actor	Student		
Trigger	Student selects an event and initiates registration		
Precondition	Student must be logged in and the event must have available tickets		
Main Flow	1. Student opens the event list 2. Student selects an event to register for 3. System checks ticket availability 4. System prompts for payment 5. Student completes payment 6. System verifies payment 7. System updates registration record 8. Student receives confirmation		
Alternate Flow – Tickets Unavailable:	3.1 System shows: "Tickets are sold out"		
Alternate Flow – Payment Failed:	6.1 System shows: "Payment unsuccessful. Please try again."		
Author	Suliman		

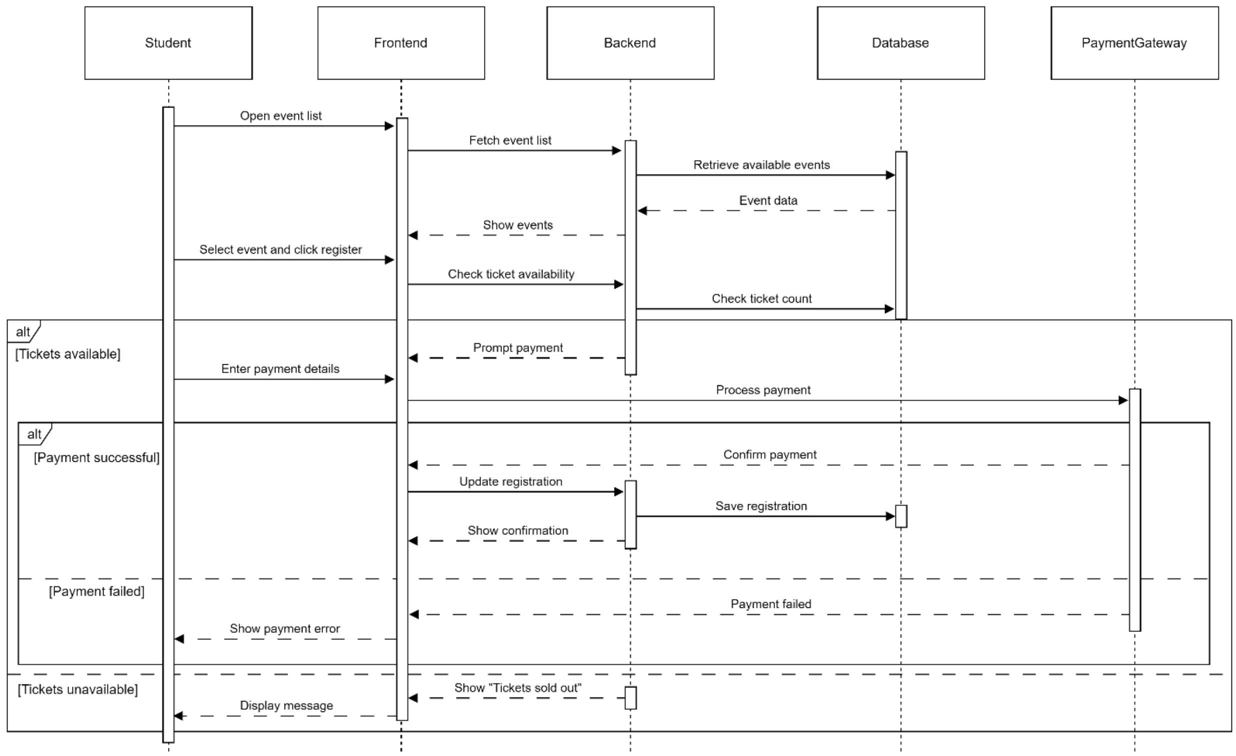

Figure 3. 3 F002 Event Registration Sequence Diagram

3.2.1.3 Event Check-In

This section defines the functional requirements and the use case specification for the Event Check-In feature of the system. It includes a detailed description of the system behavior, the primary actors involved, and the steps taken during the interaction. A sequence diagram is also provided to visually represent the process flow.

Table 3.2. 12 FR0301 Table

Requirement ID	**FR0301**	**Version**	**2.0**
Description	The system shall allow students to check in to events for which they are registered.		
Author	Suliman		

Table 3.2. 13 FR0302 Table

Requirement ID	**FR0302**	**Version**	**2.0**
Description	The system shall verify the student's registration and identity during check-in.		
Author	Suliman		

Table 3.2. 14 FR0303 Table

Requirement ID	**FR0303**	**Version**	**2.0**
Description	The system shall mark the student as checked in and log the check-in time.		
Author	Suliman		

Table 3.2. 15 FR0304 Table

Requirement ID	**FR0304**	**Version**	**2.0**
Description	The system shall prevent check-in for students who are not registered.		
Author	Suliman		

Table 3.2. 16 F003 Event Check-In Use Case Specification Table

Use Case ID	UC003	**Version**	2.0
Feature	F003 – Event Check-In		
Purpose	To allow students to check in at the event using their registration and identity		
Actor	Student		
Trigger	Student scans a QR code at the event venue		
Precondition	Student must be registered for the event and have a valid ticket		
Main Flow	1. Student arrives at the event venue 2. Student scans the QR code displayed at the check-in point 3. System verifies student registration and identity 4. If verification succeeds, the system logs check-in and records the timestamp 5. Student is allowed to enter the event		
Alternate Flow – Not Registered:	3.1 System shows: "You are not registered for this event"		
Author	Suliman		

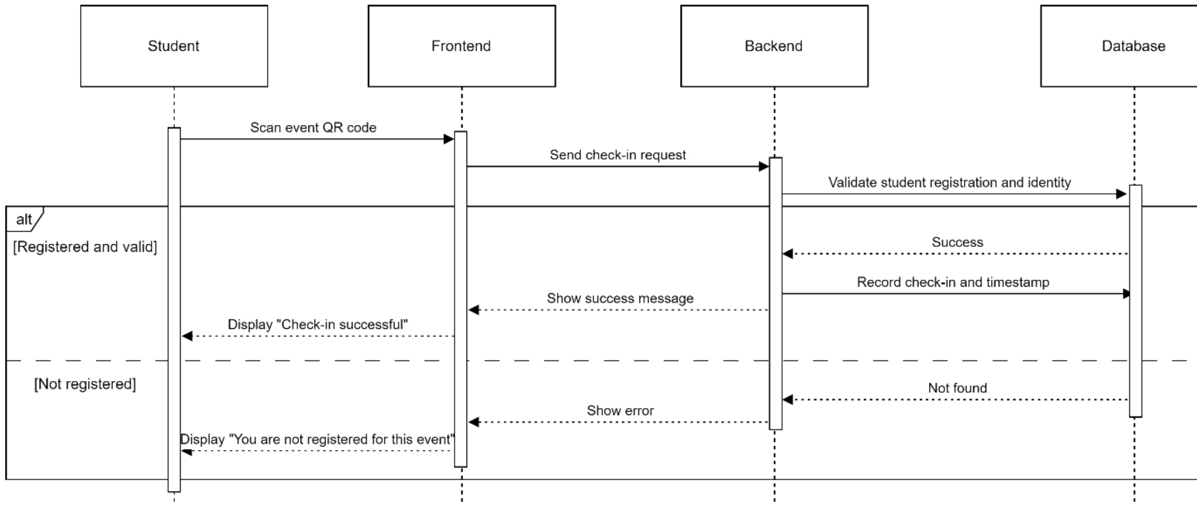

Figure 3. 4 F003 Event Check-in Sequence Diagram

3.2.1.4 Make Payment

This section defines the functional requirements and use case specification for the **Make Payment** feature of the system. In earlier versions of the SRS, this was referred to as **Process Student Payment**, primarily intended for handling payments related to **event registration**. However, payment for event tickets is now explicitly handled as part of the **Event Registration** use case and does not require separate modeling.

This **Make Payment** feature has been redefined to focus exclusively on **on-site vendor payments** during events, such as purchasing food, merchandise, or other services. This adjustment better aligns with real-time, QR-based vendor interactions. A detailed sequence diagram is provided to illustrate the updated process flow.

Table 3.2. 17 FR0401 Table

Requirement ID	**FR0401**	**Version**	**2.0**
Description	The system shall allow students to initiate payments for purchases made at event venues.		
Author	Lim Ai Nee		

Table 3.2. 18 FR0402 Table

Requirement ID	**FR0402**	**Version**	**2.0**
Description	The system shall process the payment via an integrated payment gateway.		
Author	Lim Ai Nee		

Table 3.2. 19 FR0403 Table

Requirement ID	FR0403	Version	2.0
Description	The system shall update the payment database upon successful transaction.		
Author	Lim Ai Nee		

Table 3.2. 20 FR0404 Table

Requirement ID	FR0404	Version	2.0
Description	The system shall notify students of successful or failed payment status.		
Author	Lim Ai Nee		

Table 3.2. 21 F004 Make Payment Use Case Specification Table

Use Case ID	UC004	Version	2.0
Feature	F004 – Make Payment		
Purpose	To allow students to make payments to vendors during events		
Actor	Student		
Trigger	Student selects a vendor or scans a vendor's QR code to initiate payment		
Precondition	Student must be logged in and have a valid payment method		
Main Flow	1. Student scans vendor QR code or selects vendor 2. System prompts for payment details 3. Student enters payment amount and confirms 4. System processes the payment via the gateway 5. If payment is successful, the database is updated and student is notified		
Alternate Flow – Payment Failed:	5.1 System displays: "Payment failed. Please try again."		
Author	Lim Ai Nee		

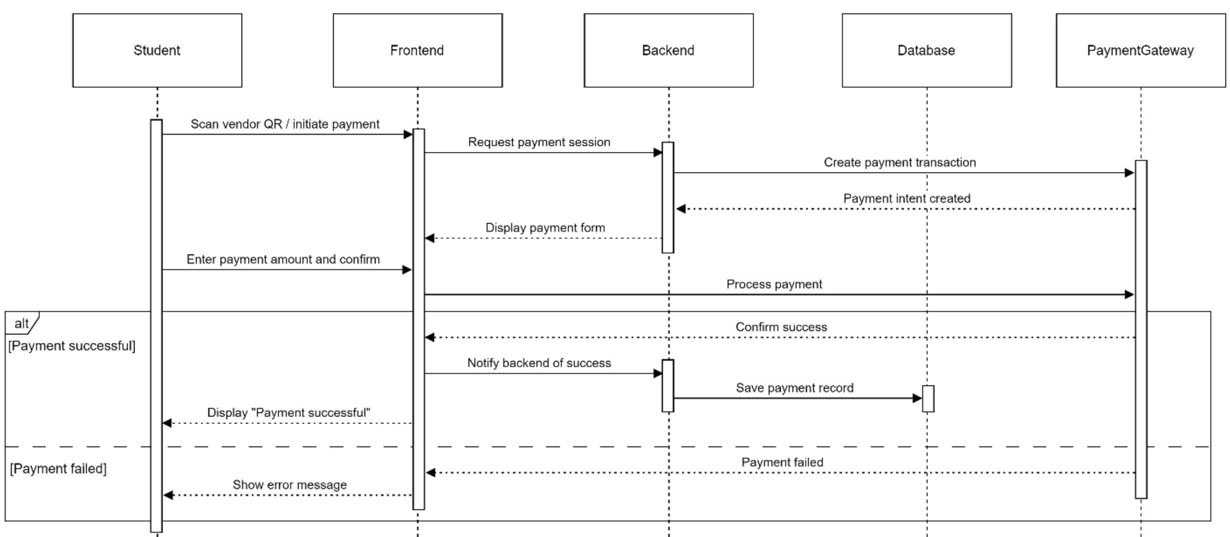

Figure 3. 5 F004 Make Payment Sequence Diagram

3.2.1.5 Manage Event

This section defines the functional requirements and use case specification for the **Manage Event** feature of the system. In the previous version of the SRS, this feature bundled all event-related actions under a single generalized flow. However, in this revised version, we explicitly represent **Create Event**, **Edit Event**, and **Delete Event** as extended use cases under **Manage Event** to more accurately reflect admin interactions and follow proper UML modeling.

A detailed sequence diagram and updated behavior flow are also provided below.

Table 3.2. 22 FR0501 Table

Requirement ID	**FR0501**	**Version**	**2.0**
Description	The system shall allow admins to access an event management interface.		
Author	Lim Ai Nee		

Table 3.2. 23 FR0502 Table

Requirement ID	**FR0502**	**Version**	**2.0**
Description	The system shall allow admins to create new events by entering required details.		
Author	Lim Ai Nee		

Table 3.2. 24 FR0503 Table

Requirement ID	**FR0503**	**Version**	**2.0**
Description	The system shall allow admins to modify existing events.		
Author	Lim Ai Nee		

Table 3.2. 25 FR0504 Table

Requirement ID	**FR0504**	**Version**	**2.0**
Description	The system shall allow admins to delete events that are no longer needed.		
Author	Lim Ai Nee		

Table 3.2. 26 FR0505 Table

Requirement ID	**FR0505**	**Version**	**2.0**
Description	The system shall validate event details before saving changes.		
Author	Lim Ai Nee		

Table 3.2. 27 FR0506 Table

Requirement ID	**FR0506**	**Version**	**2.0**
Description	The system shall update the event records in the database after each		

	operation.
Author	Lim Ai Nee

Table 3.2. 28 F005 Manage Events Use Case Specification Table

Use Case ID	UC005	**Version**	2.0
Feature	F005 – Manage Events		
Purpose	To enable admins to create, edit, or delete event records		
Actor	Admin		
Trigger	Admin opens the event management interface		
Precondition	Admin must be logged in with event management permissions		
Main Flow	1. Admin navigates to the event management section 2. System displays a list of existing events 3. Admin selects one of the following actions: • Create new event • Edit an existing event • Delete an event 4. System displays the relevant form or confirmation prompt 5. Admin submits the action 6. System validates input and saves changes to the database 7. System displays a success message		
Alternate Flow – Cancelled Action:	4.1 Admin cancels the action → return to event list		
Alternate Flow – Invalid Input:	6.1 System displays: “Please fill in all required fields”		
Author	Lim Ai Nee		

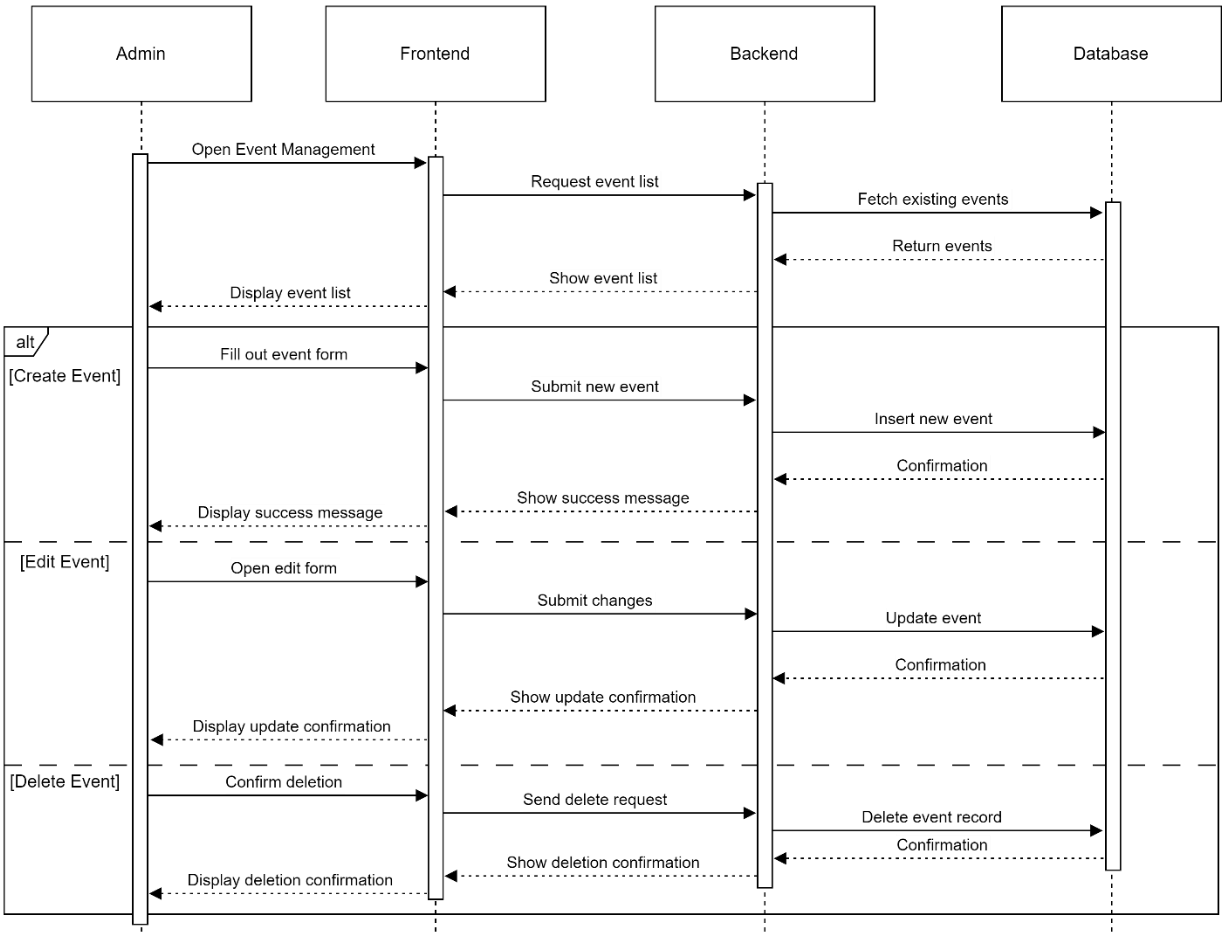

Figure 3. 6 F005 Manage Events Sequence Diagram

3.2.1.6 Manage Attendance

This section defines the functional requirements and use case specification for the **Manage Attendance** feature of the system. In the previous version of the SRS, this feature was modeled with limited detail, combining QR code generation and student list display into a simplified linear flow. In this improved version, the system flow is clearly separated into two functional interactions: **QR code generation** and **attendance list retrieval**, including database integration for real-time attendance and student identity data. A revised sequence diagram illustrates this process in line with best practices.

Table 3.2. 29 FR0601 Table

Requirement ID	FR0601	Version	2.0
Description	The system shall allow admins to generate a unique QR code for event check-in.		
Author	Azhar		

Table 3.2. 30 FR0602 Table

Requirement ID	FR0602	Version	2.0
Description	The system shall retrieve check-in logs of students for a selected event.		
Author	Azhar		

Table 3.2. 31 FR0603 Table

Requirement ID	FR0603	Version	2.0
Description	The system shall retrieve student details associated with each check-in record.		
Author	Azhar		

Table 3.2. 32 FR0604 Table

Requirement ID	FR0604	Version	2.0
Description	The system shall display the list of students who have checked in, along with their basic information.		
Author	Azhar		

Table 3.2. 33 F006 Manage Attendance Use Case Specification Table

Use Case ID	UC006	**Version**	2.0
Feature	F006 – Manage Attendance		
Purpose	To enable admins to manage student attendance during events by generating check-in QR codes and reviewing attendance records		
Actor	Admin		
Trigger	Admin accesses the attendance management interface		
Precondition	Admin must be logged in and have access to an active event		
Main Flow	1. Admin opens the attendance management tab 2. selects an event 3. System generates a unique QR code for check-in 4. Admin displays the QR code at the event 5. Admin requests the attendance list 6. System retrieves check-in logs from the event database 7. System retrieves student details from the student database 8. System displays the full list of attendees with names and check-in timestamps		
Alternate Flow – No Check-ins	6.1 System displays: “No students have checked in yet”		

Yet:	
Author	Azhar

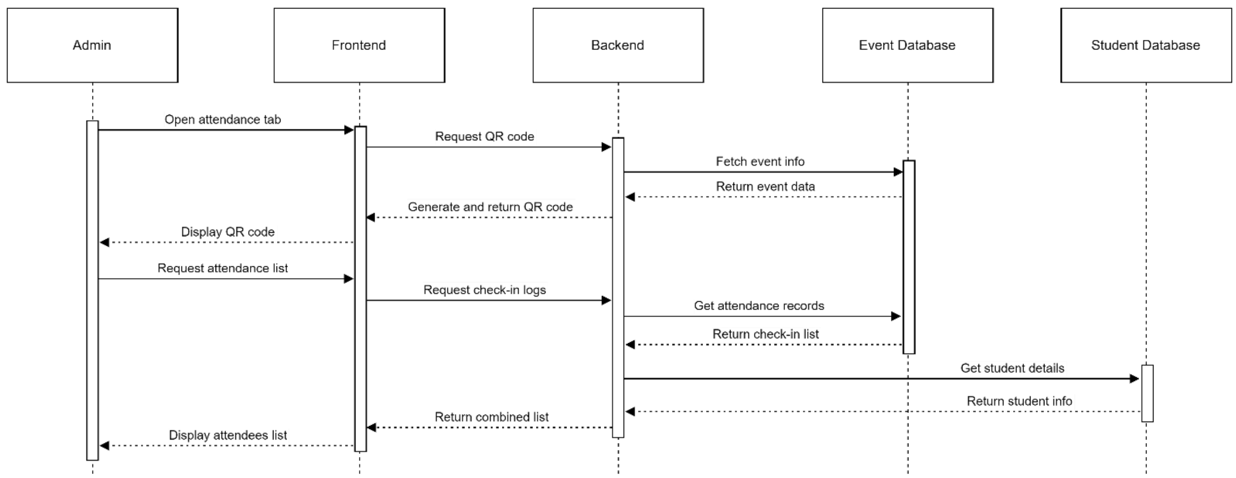

Figure 3. 7 F006 Manage Attendance Sequence Diagram

3.2.1.7 Manage Payment (Vendor)

This section defines the functional requirements and use case specification for the **Manage Payment** feature used by vendors during events. In the previous version of the SRS, the process was simplified to basic button clicks and lacked back-end interaction or structured flow. This updated version includes real-time payment entry, transaction handling, and backend database logging. The flow now separates user-facing input from data processing, and includes optional handling of change or balance returns (if applicable).

Table 3.2. 34 FR0701 Table

Requirement ID	**FR0701**	**Version**	**2.0**
Description	The system shall allow vendors to input payment amounts during transactions.		
Author	Azhar		

Table 3.2. 35 FR0702 Table

Requirement ID	**FR0702**	**Version**	**2.0**
Description	The system shall process and record each payment in the payment database.		
Author	Azhar		

Table 3.2. 36 FR0703 Table

Requirement ID	FR0703	Version	2.0
Description	The system shall confirm that a transaction has been saved before updating the interface.		
Author	Azhar		

Table 3.2. 37 FR0704 Table

Requirement ID	FR0704	Version	2.0
Description	The system shall allow vendors to mark a sale as complete and begin a new transaction.		
Author	Azhar		

Table 3.2. 38 FR0705 Table

Requirement ID	FR0705	Version	2.0
Description	The system shall display an error message if the entered amount field is empty or invalid.		
Author	Azhar		

Table 3.2. 39 F007 Vendor Payment Management Use Case Specification Table

Use Case ID	UC007	Version	2.0
Feature	F007 – Vendor Payment Management		
Purpose	To enable vendors to manage on-site payments during events		
Actor	Vendor		
Trigger	Vendor opens the in-app payment interface		
Precondition	Vendor must be logged in and registered to the event		
Main Flow	1. Vendor opens the vendor dashboard 2. Vendor clicks "Start New Sale" 3. Vendor enters the total amount to be paid 4. System records the transaction in the payment database 5. System confirms and updates the dashboard with the new payment entry 6. Vendor can start another transaction		
Alternate Flow – Invalid/Empty Field:	3.1 System displays: "Please enter a valid amount"		
Author	Azhar		

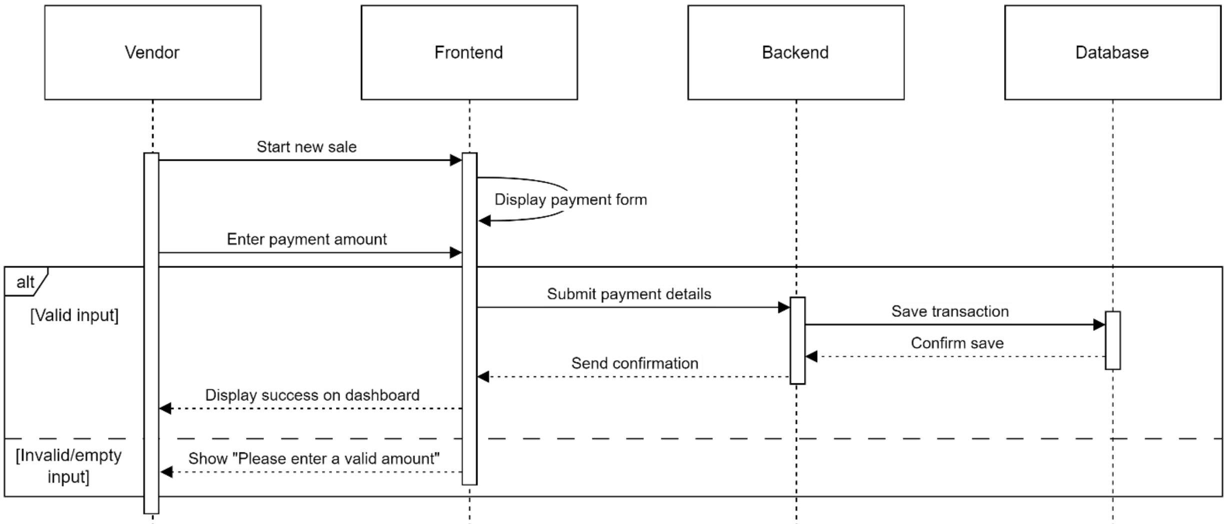

Figure 3. 8 F007 Manage Payment Sequence Diagram

3.2.1.8 View Sales Report

This section defines the functional requirements and use case specification for the **View Sales Report** feature for vendors. In the previous version of the SRS, the interaction was oversimplified and assumed the sales report was generated directly from the vendor interface without clearly involving backend verification or data retrieval. In this improved version, the flow explicitly reflects backend interaction with a unified database and provides proper error handling for empty or missing data.

Table 3.2. 40 FR0801 Table

Requirement ID	**FR0801**	**Version**	**2.0**
Description	The system shall allow vendors to request a sales report based on a defined period.		
Author	Yousef		

Table 3.2. 41 FR0802 Table

Requirement ID	**FR0802**	**Version**	**2.0**
Description	The system shall retrieve the vendor’s sales data from the database.		
Author	Yousef		

Table 3.2. 42 FR0803 Table

Requirement ID	**FR0803**	**Version**	**2.0**
Description	The system shall generate a structured report based on retrieved sales		

	records.
Author	Yousef

Table 3.2. 43 FR0804 Table

Requirement ID	**FR0804**	**Version**	**2.0**
Description	The system shall display the report to the vendor in a readable format.		
Author	Yousef		

Table 3.2. 44 FR0805 Table

Requirement ID	**FR0805**	**Version**	**2.0**
Description	The system shall show an appropriate message if no sales data is found for the selected period.		
Author	Yousef		

Table 3.2. 45 F008 View Sales Report Use Case Specification Table

Use Case ID	UC008	**Version**	2.0
Feature	F008 – View Sales Report		
Purpose	To enable vendors to view and analyze their sales activity for a specific date range		
Actor	Vendor		
Trigger	Vendor initiates a sales report request from the dashboard		
Precondition	Vendor must be logged in and have recorded transactions		
Main Flow	1. Vendor opens the report section from the dashboard 2. Vendor selects a date range and clicks “Generate Report” 3. System retrieves relevant sales data from the database 4. System generates and formats the report 5. System displays the report to the vendor		
Alternate Flow – No Sales Found:	3.1 System displays: “No sales data available for the selected period”		
Author	Yousef		

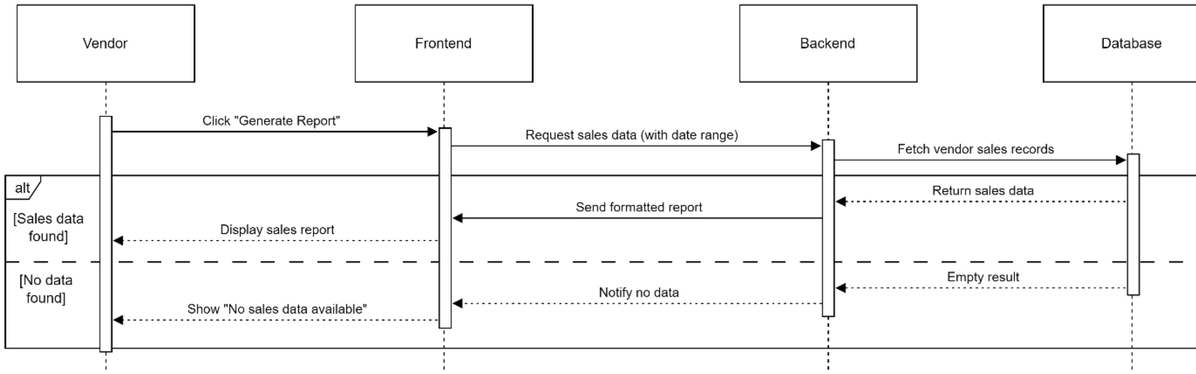

Figure 3. 9 F008 View Sales Report Sequence Diagram

3.2.1.9 Generate Event Report

This section defines the functional requirements and use case specification for the **Generate Event Report** feature. In previous versions of the SRS, event reporting functionality was implicitly handled as part of the **Manage Event** use case. In this updated version, it has been modeled as an **individual use case** to better reflect its distinct role in providing post-event analytics and documentation. This is the first dedicated version of this feature in the SRS, providing clearer structure and backend integration for generating reports.

Table 3.2. 46 FR0901 Table

Requirement ID	**FR0901**	**Version**	**1.0**
Description	The system shall allow admins to generate reports for specific events.		
Author	Yousef		

Table 3.2. 47 FR0902 Table

Requirement ID	**FR0902**	**Version**	**1.0**
Description	The system shall retrieve all relevant attendance and transaction data from the database.		
Author	Yousef		

Table 3.2. 48 FR0903 Table

Requirement ID	**FR0903**	**Version**	**1.0**
Description	The system shall compile the data into a readable and exportable format.		
Author	Yousef		

Table 3.2. 49 FR0904 Table

Requirement ID	**FR0904**	**Version**	**1.0**
Description	The system shall allow the admin to view and optionally download the		

	report.
Author	

Table 3.2. 50 FR0905 Table

Requirement ID	**FR0905**	**Version**	**1.0**
Description	The system shall notify the admin if no data is available for the selected event.		
Author	Yousef		

Table 3.2. 51 F009 Generate Event Report Use Case Specification Table

Use Case ID	UC009	**Version**	1.0
Feature	F009 – Generate Event Report		
Purpose	To allow admins to generate and access post-event reports including attendance and sales data		
Actor	Admin		
Trigger	Admin accesses the report section and selects an event		
Precondition	Admin must be logged in and the selected event must have at least one data record (attendance or transaction)		
Main Flow	1. Admin opens the reporting interface 2. Admin selects an event to generate a report for 3. System retrieves attendance and sales data related to that event 4. System formats and compiles the report 5. System displays the report and allows optional download		
Alternate Flow – No Data:	3.1 System displays: "No reportable data available for this event"		
Author	Yousef		

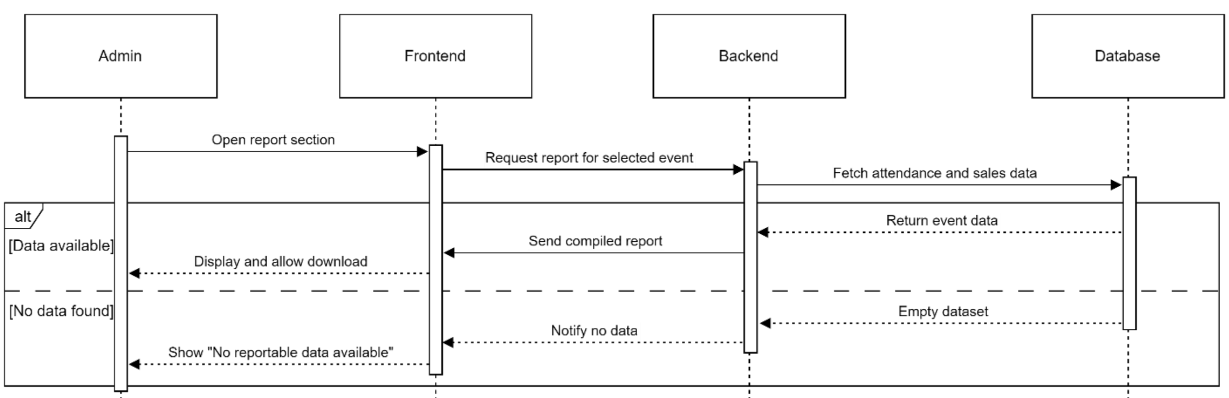

Figure 3. 10 F009 Generate Event Report Sequence Diagram

3.2.1.10 Add Vendor Account

This section outlines the functional requirements and use case specification for the **Add Vendor Account** feature. In previous iterations of the SRS, vendor account management was not explicitly defined. This updated version introduces it as a standalone feature to emphasize its importance in onboarding external service providers such as food and merchandise vendors. The feature enables administrators to register new vendor accounts through a guided form, ensuring required fields are validated, and duplicate entries are handled effectively. This dedicated modeling provides greater clarity, improves system robustness, and enhances the onboarding workflow for event-related vendors.

Table 3.2. 52 FR1001 Table

Requirement ID	**FR1001**	**Version**	**1.0**
Description	The system shall present an Add Vendor form when the administrator selects "Create Vendor".		
Author	Suliman		

Table 3.2. 53 FR1002 Table

Requirement ID	**FR1002**	**Version**	**1.0**
Description	The system shall validate all mandatory fields.		
Author	Suliman		

Table 3.2. 54 FR1003 Table

Requirement ID	**FR1003**	**Version**	**1.0**
Description	The system shall check that the vendor email address is unique in the database.		
Author	Suliman		

Table 3.2. 55 FR1004 Table

Requirement ID	**FR1004**	**Version**	**1.0**
Description	Upon successful creation the system shall store the vendor account in the vendors table.		
Author	Suliman		

Table 3.2. 56 FR1005 Table

Requirement ID	**FR1005**	**Version**	**1.0**

Description	If any validation fails, the system shall display an appropriate error message and keep the form open for correction.
Author	Suliman

Table 3.2. 57 F010 Admin Vendor Account Use Case Specification Table

Use Case ID	UC010	**Version**	1.0
Feature	F010 – Admin Vendor Account		
Purpose	Allow an admin to create a new vendor account		
Actor	Admin		
Trigger	Admin clicks “Add Vendor”.		
Precondition	Admin is authenticated		
Main Flow	1. System displays the Add Vendor form. 2. Admin fills in fields 3. System validates that mandatory fields are non-empty and email format is correct. 4. System checks that the email address is not already registered. 5. System writes the new vendor record to the. 6. System shows a success toast “Vendor created successfully; credentials sent.”		
Alternate Flow — Invalid Input	3.1 If any required field is empty or email invalid, system highlights the field(s) and shows “Please complete all mandatory entries with valid data.”		
Alternate Flow — Duplicate Email	4.1 If the email already exists, system displays “This email is already registered” and prevents submission.		
Author	Suliman		

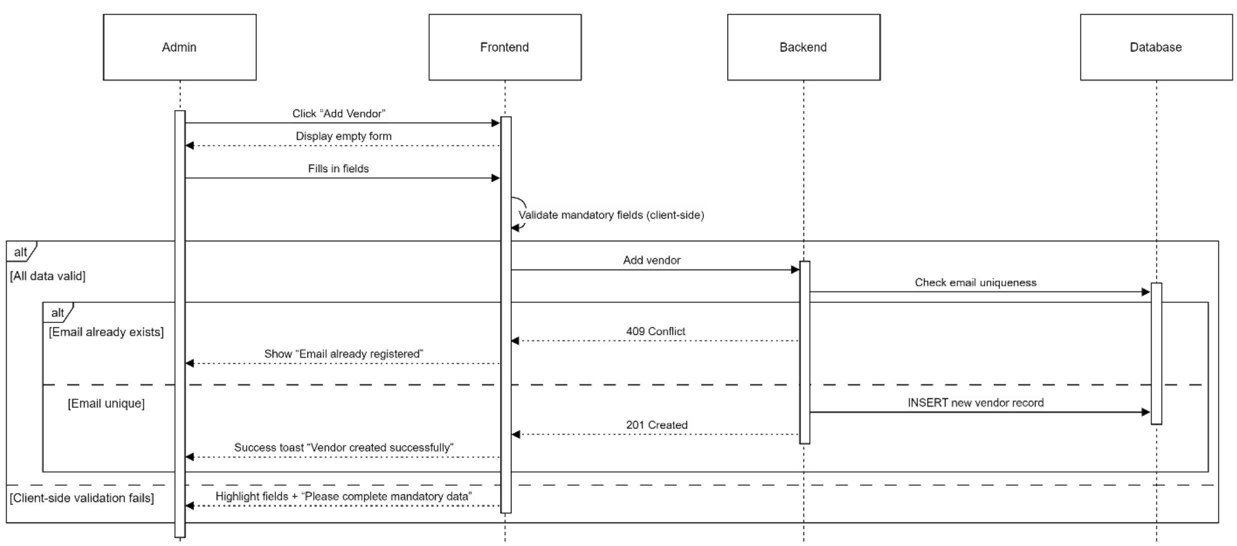

Figure 3. 11 F010 Add Vendor Account Sequence Diagram

3.3 Performance Requirements

Table 3.3. 1 describes the performance requirements and their descriptions for the Campus Event Check-in System. These requirements ensure that the system performs efficiently under various conditions, including handling concurrent users, ensuring quick response times for check-in and transactions, supporting system scalability, and maintaining data synchronization across devices.

Table 3.3. 1 Performance Requirements Table

Requirement ID	*Description*	*Priority*	*Author*
REQ_P001	The system must support up to 300 concurrent users without any noticeable performance degradation.	High	Lim Ai Nee
REQ_P002	Check-in validation and ticket verification must complete in under 5 seconds.	High	Lim Ai Nee
REQ_P003	All on-site transactions (e.g., purchases from vendors) must be processed accurately and efficiently.	High	Lim Ai Nee
REQ_P004	Upon successful registration, the event ticket confirmation should be issued within 10 seconds of form submission.	Medium	Lim Ai Nee
REQ_P005	The system must scale to support up to 3,000 total users, maintaining stability and preventing data loss.	Medium	Lim Ai Nee
REQ_P006	Any changes made by admins (such as event updates or ticket changes) must synchronize across all devices.	Medium	Lim Ai Nee

3.4 Usability Requirements

Table 3.4. 1 outlines the usability requirements for the Campus Event Check-in System. These requirements focus on delivering an intuitive and accessible user experience across both web and mobile platforms. Emphasis is placed on interface clarity, error recovery, layout organization, and support for diverse user roles to ensure that all users can interact with the system confidently and efficiently.

Table 3.4. 1 Usability Requirements Table

Requirement ID	*Description*	*Priority*	*Author*
REQ_UR001	The system must deliver a user-friendly interface accessible on both web and mobile platforms.	High	Lim Ai Nee
REQ_UR002	All users (students, vendors, and admins) must be able to complete their tasks (e.g., check-in, view sales, register for events) without confusion or error.	Medium	Lim Ai Nee
REQ_UR003	The user interface should maintain readable fonts, well-organized layouts, and support keyboard navigation to ensure accessibility.	High	Lim Ai Nee
REQ_UR004	Error messages must be clear and informative, providing users with the guidance to recover from errors independently.	High	Lim Ai Nee

3.5 Logical Database Requirements

Table 3.5. 1 presents the logical database requirements for the Campus Event Check-in System. Each entry describes a core system database and its function in storing, managing, and linking key data elements such as event details, ticket records, attendance logs, and transaction histories. These databases support the backend processes necessary for efficient event management, user tracking, and secure payment handling.

Table 3.5. 1 Logical Database Requirements Table

Requirement ID	*Table Name*	*Revised Description*	*Author*
DB_001	Events	Maintains metadata for each campus event including name, date, and venue.	Azhar
DB_002	Tickets	Links registered users to specific events through unique ticket records.	Azhar
DB_003	Checkins	Logs each check-in instance during the event for attendance tracking.	Azhar
DB_004	Transactions	Records all on-site vendor transactions and payment activities.	Azhar
DB_0005	Vendors	Records the vendors accounts	Suliman

3.6 Design Constraints

Table 3.6. 1 outlines the design constraints that impact the development and deployment of the Campus Event Check-in System. These constraints include technical, operational, and regulatory limitations such as device compatibility, reliance on third-party systems, internet connectivity

requirements, excluded features, and compliance with data protection regulations. Recognizing these constraints is essential for setting realistic expectations and ensuring system feasibility within the defined scope.

Table 3.6. 1 Design Constraints Table

Requirement ID	*Constraint*	*Revised Description*	*Author*
CON_001	Device Compatibility	System performance may degrade on outdated devices or unsupported operating systems.	Azhar
CON_002	Third-party Integrations	The system depends on external services, such as university databases and a secure payment gateway, for full functionality.	Azhar
CON_003	Network Dependence	A stable internet connection is required for real-time operations like check-ins and payment logging. Offline usage is not supported.	Azhar
CON_004	Limited Scope	Features like venue navigation, mapping, and social media integration are outside the scope of this release.	Azhar
CON_005	Security Compliance	The system must adhere to data protection regulations when handling user credentials, financial transactions, and personal data.	Azhar

3.7 Software System Attributes

Table 3.7. 1 summarizes the essential software system attributes required for the Campus Event Check-in System. These attributes define the non-functional qualities of the system, such as reliability, availability, security, maintainability, and portability. Each attribute ensures that the system performs effectively under various conditions while remaining secure, scalable, and accessible across multiple platforms and usage scenarios.

Table 3.7. 1 Software System Attributes

Requirement ID	*Attribute*	*Rephrased Description*	*Author*
ATTR_001	Reliability	The system shall auto-recover from crashes within one minute, ensuring no data loss. All critical data such as check-ins and transactions will be saved and restored automatically.	Yousef
ATTR_002	Availability	The application shall maintain uptime throughout event hours, with auto-recovery features restoring the latest saved state.	Yousef
ATTR_003	Security	TLS encryption will be used for data transmission, and AES encryption will secure data at rest. Integrity checks will safeguard against tampering and unauthorized access.	Yousef

ATTR_004	Maintainability	The system shall be modular and well-documented, enabling easy maintenance, bug fixes, and future enhancements.	Yousef
ATTR_005	Portability	The application shall be compatible with both Android and iOS platforms.	Yousef

3.8 Supporting Information

Table 3.8. 1 provides supporting information relevant to the development and deployment of the Campus Event Check-in System. It includes documentation plans, source code management practices, legal and ethical compliance considerations, and the tools and frameworks selected for implementation. These elements ensure that the system is well-documented, legally compliant, collaboratively developed, and built using appropriate technologies aligned with project needs.

Table 3.8. 1 Supporting Information (SRS1)

Requirement ID	*Category*	*Rephrased Description*	*Author*
INFO_001	Documentation	- User guides for students, vendors, and admins - Backend API documentation for third-party integration	Yousef
INFO_002	Source Code Management	- The project source code will be hosted on GitHub for version control and collaboration	Yousef
INFO_003	Legal and Ethical Compliance	- Data collection will comply with university privacy policies and local legal requirements - Payment processing will follow PCI DSS standards via a secure gateway	Yousef
INFO_004	Tools & Frameworks	- The app will be developed using Flutter; the backend database will use either SQLite or MongoDB depending on deployment requirements.	Yousef

3.8.1 Validation Activities

This section will contain information about the validation activities the team have applied in order to check and validate the SRS report if it meets the three requirements engineering dimensions. However, there are two sub-sections in this section, the first one will show the validation session the team had conducted. The second sub-section will show the validation details.

3.8.1.1 Validation Sessions

Review Object
ID / Title / Version: v1.0
Author / Groupe Name: G1
Meeting Information
Date / Time: 8 / 6 / 2025 3:00 pm
Participation (Roles):

Suliman (Inspector, Minute-taker)
Yousef (Moderator, Inspector, Reader)
Azhar (Author, Stakeholder)
Lim Ai Nee (Organizer, Inspector, Stakeholder)

Review Object
ID / Title / Version: v1.0
Author / Groupe Name: G1
Meeting Information
Date / Time: 8 / 6 / 2025 3:00 pm
Participation (Roles):

Suliman (Moderator, Inspector, Reader)
Yousef (Inspector, Minute-taker)
Azhar (Author, Stakeholder)
Lim Ai Nee (Organizer, Inspector, Stakeholder)

Review Object
ID / Title / Version: v1.0
Author / Groupe Name: G1
Meeting Information
Date / Time: 16 / 6 / 2025 3:00 pm

Participation (Roles):

Suliman (Moderator, Inspector, Reader)

Yousef (Inspector, Minute-taker)

Azhar (Organizer, Inspector, Stakeholder)

Lim Ai Nee (Author, Stakeholder)

3.8.1.2 Validation Details

This subsection presents two artefacts:

- **Table 3.8. 2 Defects Table** – lists every defect discovered during validation. Each entry shows the page or use-case ID where the defect appears, the reviewer who raised it, its impact, and the team's recommended action. Defects are prefixed **C-**, **D-**, or **A-** to identify Content, Documentation, and Agreement issues respectively.

The "ID / Page No" column always references either a use-case number (e.g., **UC010**) or the chapter/sub-chapter where the issue existed in the previous SRS draft, allowing traceability across revisions.

Table 3.8. 2 Defects Table

No	ID / Page No	Defect Description	Type	Inspector Name	Impact	Recommended Action
C-1	1.3.3	Vendor account source is undefined; no onboarding workflow	Agreement Defect + Conflict	Suliman	System assumes vendor accounts exist yet provides no creation flow	Add a mechanism to add vendors account
C-2	UC008, UC006	Several sequence diagrams lack alternate/error flows (e.g., Sales Report, Admin actions)	Content Defect	Suliman	Edge-case behavior undocumented	Update all UC tables with error flows
C-3	UC005, UC006, UC007, UC008	Sequence diagrams omit backend/database calls (e.g., ticket update, payment record)	Content Defect	Suliman	Data-consistency path invisible	Update diagrams
C-4	3.5	Duplicate local **Users/Roles** tables while identity is handled by university DB	Content Defect	Yousef	Risk of data divergence	Remove duplicate tables
C-5	UC005, UC007…	"Payment DB / Ticket DB" labels imply separate physical DBs	Content Defect	Azhar	Mis-leads architecture readers	Rename to logical tables in a single DB
C-6	1.3	The system is missing the Goals	Content Defect	Suliman	System purpose and high-level objectives are unclear; traceability to goals impossible	Add "System Goals" subsection (list measurable, high-level goals) under 1.3
D-1	Most Tables	Inconsistent table formatting & captions	Formatting Defect	Yousef	Hard to scan / reference	Update all tables with a standardized styling
D-2	1.1, 1.3.2	Mixed terminology: "Student ID" vs. "University ID"	Language Defect	Lim Ai Nee	Causes confusion	Choose one term
D-3	UC003	Grammar issues in flow descriptions ("Student arrive", etc.)	Language Defect	Azhar	Reduces professionalism	Correct sentences

D-4	Most Figures	Figure numbers/captions missing or duplicated	Structural Defect	Azhar	Navigation difficult	Unique numbering
D-5	UC003, UC005	Use-case flows mix bullets, numbers, vague labels	Formatting Defect	Suliman	Inconsistent readability	Re-format flows
D-6	UC007	Sequence diagrams break UML rules (messages from actors, 2 actors on 1 diagram)	Modeling Defect	Yousef	Non-standard diagrams	Redraw correctly
D-7	3.4.4	References to SQLite in API text but DB undecided	Language Defect	Lim Ai Nee	Creates false specificity	Edit Software Interface section and generalize wording
D-8	3.4.2	UI specification supplied only as mock-ups; interface-element tables missing	Documentation Defect	Yousef	Hard for developers/testers to see precise screen elements	Provide detailed UI tables per function
D-9		Use-case diagram shows erroneous subsystem boundary	Modeling Defect	Azhar	Misrepresents system scope	Replace diagram with corrected single-system use-case
D-10		Interface Requirements section located in wrong chapter	Structural Defect	Lim Ai Nee	Document structure deviates from ISO/IEC 29148	Relocate section to 1.3 Product Perspective
A-1		UI shows only mobile screens; laptops/tablets promised for admins & vendors	Agreement Defect	Azhar	Platform mismatch	Clarify responsive design
A-2		Stripe named elsewhere but not in System Interfaces table	Agreement Defect	Yousef	Integration uncertainty	Update Sys Interfaces row

3.8.2 *Conflict Analysis & Resolution*

Table 3.8. 3 Conflicts Table – captures items that could not be classified as straightforward defects because reviewers initially disagreed on how to resolve them. For each conflict we record the description, the rationale behind the disagreement, and the final resolution reached after negotiation.

Table 3.8. 3 Conflicts Table

No	Conflict Description	Conflict Analysis (reason for disagreement)	Final Resolution (what was changed)
1	Vendor onboarding undefined – some reviewers assumed self-registration via university DB; others expected admin-created accounts	System design only authenticates students via university DB; vendors are external. Expectations differed.	Resolved: Added Admin Adds Vendor Account flow (UC-010) and UI tables; clarified in System Interfaces that vendors are created locally.
2	Screens showed only mobile layouts while admins/vendors expected tablet / laptop use	Mobile-only mock-ups implied single-platform support.	Resolved: Added statement that UI is fully responsive.
3	Stripe gateway referenced in text but missing in table	Omission created uncertainty about payment integration.	Resolved: Stripe added to 3.4.1 System Interfaces row (Payment Gateway Integration).
4	External Interface Requirements located in wrong chapter	ISO/IEC 29148 requires interface desc. In 1.x context.	Resolved: Section relocated to 1.3; cross-refs updated.

3.8.3 Traceability Matrix

Table 3.8. 4 presents the Traceability Matrix, which maps functional requirement scenarios (REQ_Fxxx) to the corresponding system goals (REQ_Gxxx). This matrix ensures that all defined goals are adequately addressed by specific requirements, helping maintain alignment between stakeholder objectives and system functionality throughout the development process. It supports verification, validation, and completeness of the requirement coverage.

Table 3.8. 4 Traceability Matrix Table

Scenarios	**REQ_GO01**	**REQ_GO02**	**REQ_GO03**	**REQ_GO04**	**REQ_GO05**	**REQ_GO06**	**REQ_GO07**
REQ_F001					satisfies	satisfies	
REQ_F002				satisfies	satisfies		
REQ_F003	satisfies			satisfies	satisfies		
REQ_F004		satisfies				satisfies	
REQ_F005			satisfies				satisfies
REQ_F006			satisfies	satisfies			satisfies
REQ_F007		satisfies	satisfies			satisfies	
REQ_F008			satisfies				satisfies
REQ_F009			satisfies	satisfies			satisfies
REQ_F010		satisfies				satisfies	

4. VERIFICATION

4.1 Verification Approach

HOW

Verification will be performed through **user, integration, and system testing**. Test cases will be derived directly from SRS requirements to maintain traceability.

WHO

- *Developers* carry out integration and system tests during development.
- *End-users* (students, admins, vendors) participate in user-acceptance testing.
- The *project manager* oversees the process and signs off when all criteria are met.

WHEN

- **Integration testing** begins once components are integrated.
- **System testing** is executed after all modules are in place.
- **User testing** occurs in the pre-deployment phase.
- Final verification is completed before the system goes live.

WHERE

- Integration and system tests run on university-hosted development / test servers.
- Verification artefacts (test plans, results) are stored in the shared GitHub repository, accessible to all stakeholders.

4.2 Verification Criteria

Table 4. 1 outlines the verification criteria for the Campus Event Check-in & Payment System. Each requirement area is paired with specific, measurable conditions to determine successful implementation and system behavior. These criteria ensure that key functions—such as login, ticketing, purchases, and admin updates—perform within defined time limits and meet expectations for accuracy, scalability, accessibility, and reliability.

Table 4. 1 Verification Criteria

Requirement Area	Verification Criteria
Login	Users (students, admins, vendors) must log in with valid credentials. Response time ≤ 3 s.
Event Registration	Students must register and receive a ticket within 6 s of submission.
Ticket Verification	System must validate a ticket within 2 s.
On-site Purchases	Vendors must complete a transaction ≤ 3 s after payment confirmation.
Data Consistency	All transactions and check-ins are recorded reliably in the database.
Error Handling	On failure, the system must return an error message ≤ 2 s.
Accessibility	UI meets accessibility standards (readable fonts, keyboard navigation).
Load Handling	System functions with up to **300 concurrent** users (no page load > 3 s).
Scalability	Supports scaling to **3 000 total** users without errors.
Data Synchronisation	Admin changes (e.g., event status) propagate to all clients within 10 s.
Timezone & Clock Sync	All timestamps stored in UTC and displayed in MYT (GMT+8).
UI Response Time	Any UI action (navigation, form submission) triggers feedback ≤ 3 s.

5. APPENDICES

5.1 Assumptions and Dependencies

Assumptions

- The university provides a secure API or direct connection to its student-identification database.
- Event venues have reliable Internet connectivity to support real-time check-ins and payments.
- Students possess valid IDs and tickets at check-in.
- Students, admins, and vendors have Internet access (campus Wi-Fi or mobile data).
- Administrators operating the system are trained to use the platform effectively.

Dependencies

- Appropriate device compatibility for QR check-in (camera availability, troubleshooting).
- Integration with the university student-ID database for real-time identity verification.
- Integration with a PCI-DSS-compliant payment gateway for transaction processing.

5.2 Acronyms and Abbreviations

Table 5. 1 lists the acronyms and abbreviations used throughout this document, along with their corresponding definitions, to ensure clarity and consistency for all readers.

Table 5. 1 Acronyms and Abbreviations

Acronym	Definition
API	Application Programming Interface
e.g.	For example
GMT+8	Greenwich Mean Time + 8 hours
MYT	Malaysian Time
TLS	Transport Layer Security
AES	Advanced Encryption Standard
PCI	Payment Card Industry (as in PCI-DSS)
HTTPS	Hypertext Transfer Protocol Secure